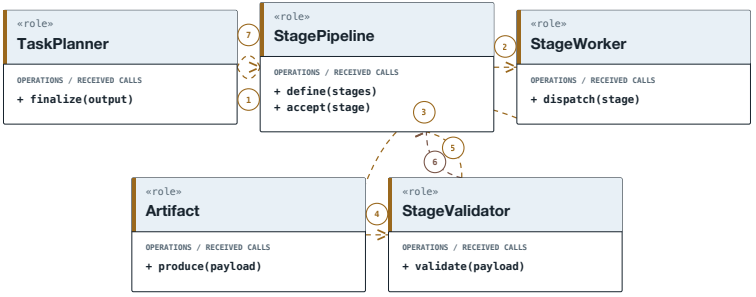
W01 Prompt chaining

Workflow / UML CLASS VIEW / ANALYTICAL MODEL

Pass intermediate results through planned stages.

UML CLASS VIEW

dashed arrow: call dependency · brown: return/failure · analytical, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	A task is decomposed into ordered agent or prompt stages.
TRIGGER	One stage produces an intermediate result consumed by the next.
STRUCTURE	Stage graph, typed handoff payload, validation, and terminal output.
EFFECT	Stages execute in order and downstream input is attributable to upstream output.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Stage order and handoff schema are defined.
ENSURES	Every stage receives validated input and the final output identifies its stage lineage.
INVARIANT	A later stage cannot execute before its prerequisites complete.
VIOLATION	A stage failure stops, retries, or branches according to declared policy.

RELATIONSHIPS / ANALYTICAL CALLS

number · caller → callee · message

1 TaskPlanner → StagePipeline define(stages)
2 StagePipeline → StageWorker dispatch(stage)
3 StageWorker → Artifact produce(payload)
4 Artifact → StageValidator validate(payload)
5 StageValidator → StagePipeline accept(stage)
6 StageValidator ↩ StagePipeline retry(error)
7 StagePipeline → TaskPlanner finalize(output)

VARIATION / TEST / NEGATIVE CONTROL

VARIATION	Sequential agents, prompt templates, pipelines, and transform nodes.
TEST_OBLIGATION	Test order, input propagation, stage failure, and branching.
NEGATIVE_CONTROL	Independent prompts with no intermediate-result dependency do not count.